Master's degree in biomedical engineering

Academic year 2025/2026

MEDICAL IMAGE PROCESSING

# Retinal Vessel Challenge


GROUP BC7

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# 1. INTRODUCTION

Retinal vessels segmentation aims to separate the different retinal vasculature structure tissues from the background of fundus images and from other anatomical retinal structures such as the optic disc, macula, and possible lesions. In recent years, this task has attracted increasing attention due to the widespread use of non-invasive fundus imaging techniques and the crucial information contained in the retinal vasculature.

Indeed, the retinal vascular structure serves as diagnostic characteristic for several disorders such as diabetic retinopathy, glaucoma, hypertension, coronary artery disease, and atherosclerosis. Traditionally, retinal vessel analysis is performed manually by ophthalmologists, which is not only time-consuming but also prone to human error and fatigue, potentially leading to inconsistent or inaccurate assessments.

For these reasons, retinal vessel segmentation has become a fundamental step in the automated analysis of fundus images. [1] [2]

The objective of this project is to design algorithms capable of segmenting retinal vessels from color fundus images.

The following sections will discuss the design choices that led to the final strategy, which was implemented in Python using the Google Colab environment, as illustrated in Figure 1.

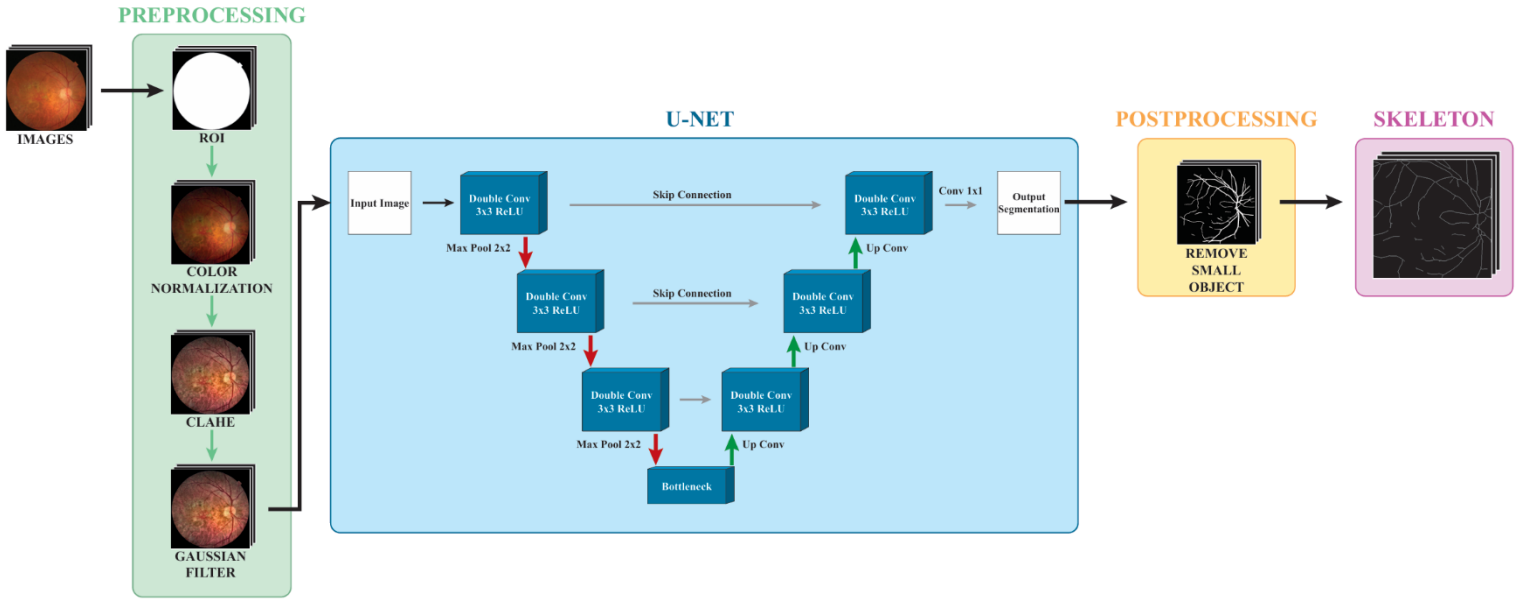



*Figure 1:Pipeline*

# 2. MATERIALS AND METHODS

## 2.1 Dataset

The dataset includes 600 images provided in RGB PNG format with uint8 encoding. Each image is associated with a manually generated segmentation mask created by expert operators. The masks follow the same format and encoding as the original images and adopt a binary representation, in which 0 indicates background pixels and 255 indicates vessel pixels.

The dataset is characterized by significant visual heterogeneity, presenting variations in chromatic variability as shown in Figure 2.

*Figure 2: visual heterogeneity between images of the dataset*

## 2.2 Metrics for evaluating results

To evaluate the alignment between the predicted and actual masks and assess various aspects of segmentation performance, it is necessary to choose some evaluation metrics. In the retinal vessel segmentation process, the outcome is a pixel-based classification result: any pixel is classified either as vessel or surrounding tissue. Consequently, there are four possibilities: true positive (TP) where a pixel is identified as vessel in both the ground truth and segmented image, true negative (TN) where a pixel is classified as a non-vessel in the ground truth and the segmented image, false negative (FN) where a pixel is classified as non-vessel in the segmented image but as a vessel pixel in the ground truth image, and false positive (FP) where a pixel is marked as vessel in the segmented image but non-vessel in the ground truth image. All quantitative performance metrics are reported as Mean ± Standard Deviation across the dataset to demonstrate both the average performance and the variability of the model.

In addition to Dice and clDice, the following metrics were evaluated:

- Matthews Correlation Coefficient (MCC)

MCC (1) is a key evaluation metric due to its robustness in handling class imbalance, which is a common challenge in retinal vessel segmentation, where foreground (vessels) occupies a small portion of the image compared to the background. MCC considers TP, TN, FP, and FN, making it a more reliable indicator of overall model performance. By capturing both sensitivity and specificity in a single value ranging from -1 to +1, MCC offers a balanced and interpretable measure of classification quality. MCC is the only binary classification rate that generates a high score only if the binary predictor was able to correctly predict the majority of positive data instances and the majority of negative data instances [3]

$$MCC = \frac{(TP \cdot TN) - (FP \cdot FN)}{\sqrt{[(TP+FP)(TP+FN)(TN+FP)(TN+FN)]}} \quad (1)$$

- Intersection over Union (IoU)

It ranges from 0 to 1. It measures the statistical overlap between the predicted set, and the ground truth set as shown in (2). Compared to the Dice coefficient, IoU penalizes more over-segmentation and under-segmentation.

$$IoU = \frac{TP}{TP+FP+FN} \quad (2)$$

- Recall

Recall is the ratio of TP to the sum of TP and FN, representing the probability of correct predictions among all positive classes (3). A Recall score of 1 indicates the total absence of FN meaning no part of the object was missed, so in segmentation tasks serves as a direct indicator of undersegmentation.

$$Recall = \frac{TP}{TP+FN} \tag{3}$$

- Precision

Precision (4) indicates the probability that the prediction is correct among all the predicted positive samples. A Precision score of 1 implies that the model produced no FP. In the specific context of segmentation, Precision serves as a critical indicator of oversegmentation.

$$Precision = \frac{TP}{TP+FP} \tag{4}$$

- Accuracy

Accuracy (5) represents the ratio of correctly classified observations to the total observations. From a statistical perspective, in an imbalanced dataset like retinal imagery where background pixels dominate, Accuracy is significantly biased towards the majority class. Therefore, it is reported mainly for completeness but is not treated as a primary indicator of segmentation performance.

$$Accuracy = \frac{TP+TN}{(TP+TN+FP+FN)} \tag{5}$$

# 2.3 Dataset division

Starting from the provided datasets, the number of items was reduced by 10% from each dataset to create the test set: in particular are selected 50 images from the given training set and 10 images from the given validation set. Therefore, the pipeline is applied and evaluated on the following image subdivision:

• TRAINING SET: 450 images

• VALIDATION SET: 90 images

• TEST SET: 60 images

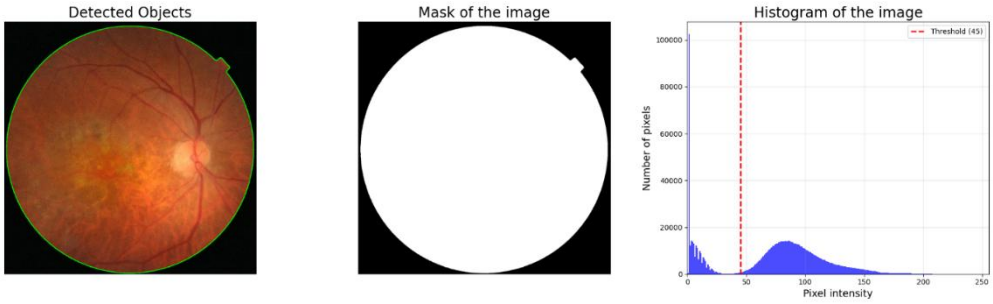
# 2.4 Pre-processing

## 2.4.1 REGION OF INTEREST (ROI)

The fundus images may exhibit variations in the background across the image. Mask generation is the process of selecting the Region of Interest (ROI); it labels the pixels of the retinal fundus within the entire image and excludes the background.

The ROI is defined using a segmentation strategy that implies a temporary conversion of the image in grayscale. Adapting the method proposed by Abdullah et al. [4] for the optic disc detection, the Hough Circle Transform is employed to identify the primary circular boundary (radius range: 200–700 pixels), utilizing a temporarily smoothed input (median blur, kernel size 5x5) to enhance geometric detection. However, since the retinal boundary is not perfectly circular, specifically due to a square marker on the border, geometric modeling alone is insufficient.
To preserve these non-circular features, a supplementary intensity mask is generated. The optimal threshold is selected based on the analysis of the pixel intensity histogram. As shown in the histogram distribution of Figure 3, this value is in the valley separating the dominant background peak from the retinal signal, ensuring the exclusion of background noise while retaining valid retinal pixels. This threshold is applied directly to the image. The final ROI is obtained via a bitwise OR operation between the geometric mask and the intensity mask. To isolate the retinal area, a bitwise AND

operation is applied between the original image and the computed binary mask. This step effectively suppresses all background content by setting the pixel intensities outside the ROI to zero, while preserving the original chromatic information within the retinal fundus.
In the color normalization stage processing is strictly confined to the ROI. The mask is then reapplied at the end of preprocessing stage ensure a perfectly uniform background.




*(a)* *(b)* *(c)*

*Figure 3: Retinal ROI extraction process. (a) Original fundus image with the detected retinal boundary superimposed in green. (b) Resulting binary mask highlighting the Field of View (FOV). (c) Pixel intensity histogram of the original image used to determine the optimal segmentation threshold (indicated by the red dashed line).*

## 2.4.2 COLOR NORMALIZATION

Given that the dataset is characterized by significant visual heterogeneity and substantial chromatic variability, a preprocessing step is required to standardize the color distribution.
Following the methodology proposed by Sweidan et al. [5], the images are first converted from RGB to the LAB color space to decouple luminance from chromatic information. The LAB color space provides more uniform color differences with respect to human visual perception. It consists of three orthogonal components: luminosity L (from dark to bright), chromaticity A (from green to red), and chromaticity B (from blue to yellow). For each color channel across the entire training dataset, a global statistical target is established by computing the mean (target mean) and standard deviation (target standard deviation) of the pixel values. These statistics are calculated exclusively within the active retinal area defined by the binary masks, ensuring that the black background did not bias the distribution. Each image in the dataset is then normalized by shifting and scaling its local intensity distribution to match these global target parameters, resulting in a harmonized dataset with consistent contrast and color characteristics (Figure 4). Global parameters values are: target mean=[74.57, 151.81, 158.29]; target standard deviation= [34.50, 14.12, 14.01].

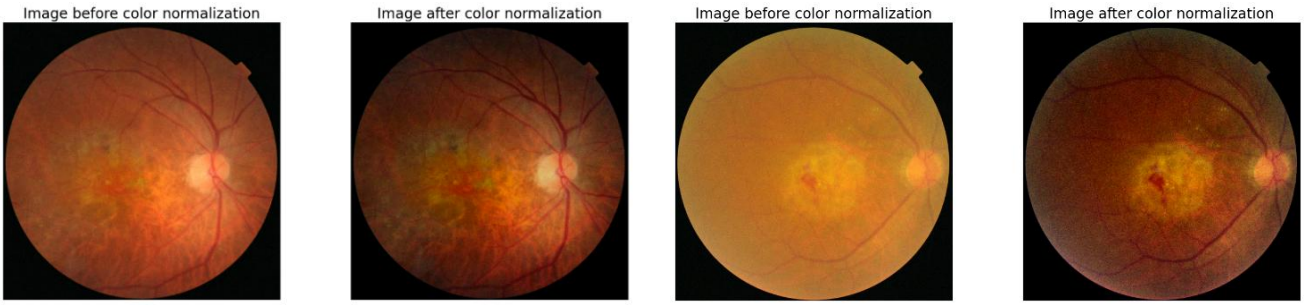



*Figure 4: Results of the color normalization process on two sample images*

## 2.4.3 CONTRAST LIMITED ADAPTIVE HISTOGRAM EQUALIZATION (CLAHE)

To enhance the contrast, CLAHE is applied (Figure 5). The primary objective is to maximize the separation between the vascular structures and the background, enabling the neural network to more effectively detect vessels. This method divides the image into small regions called tiles; the histogram on each tile is equalized so that local contrast is enhanced. This local enhancement technique can result in noise dominating in individual local regions. Hence, a clip limitation strategy is employed

to prevent local contrast from hitting the maximum. In our method, the number of tiles is 16 × 16, and the clip limit is 2.5. The evaluation of these parameters is presented in Table 2 and in Table 5.

Based on Zhou et al. [6] CLAHE is applied only to the L channel. By using the LAB color space as described above, we can enhance the local contrast without affecting the color information. After this enhancement, the image is converted back to the RGB color space for the next steps of the process.

*Figure 5: Contrast enhancement using CLAHE*

### 2.4.4 GAUSSIAN FILTER

After the CLAHE processing, a Gaussian smoothing filter is applied (Figure 6). This step is necessary to reduce high-frequency noise that could degrade the efficacy of the following segmentation algorithms. The smoothing operation was implemented utilizing a kernel size of 3x3 with a standard deviation of 0.5.

This specific configuration was determined through an optimization process. After evaluating various filter settings, comparative analysis demonstrated that this parameter set maximized the overall segmentation performance of the neural network.

Furthermore, the choice of the Gaussian kernel itself was validated against other smoothing methods, such as median and averaging filters, which proved less effective for this specific task. The numerical evidence that guided the selection of both the filter type and the parameters is reported in the Results section, in Table 3.

*Figure 6: Effect of Gaussian smoothing*

## 2.5 Neural Network

The adopted model is implemented using UNet, a convolutional neural network architecture designed for image analysis tasks. The U-shaped architecture of the network helps to maintain a higher-resolution feature map throughout the network that allows strong performance in medical segmentation tasks.

The UNet used is composed of three encoder blocks, one bottleneck block, and three decoder blocks, resulting in a symmetric encoder–decoder architecture with skip connections between corresponding levels. The network is designed to process images with dimensions of 1024x1024x3 and the corresponding binary masks with the same dimensions (1024x1024). Two classes are considered, corresponding to background and vessel. The network parameters set are: initial filters = 32; depth=3; number of epochs = 20; batch size = 4; learning rate = 1e-3; Loss Function=BCEWithLogitsLoss; optimizer = ADAM.

The chosen neural network is a non-pre-trained U-Net so different combinations of hyperparameter, are tested to optimize its performance. In particular, to increase the batch size, a resize of the images to 512x512 is attempted, but the performance obtained is worse (Table 4).

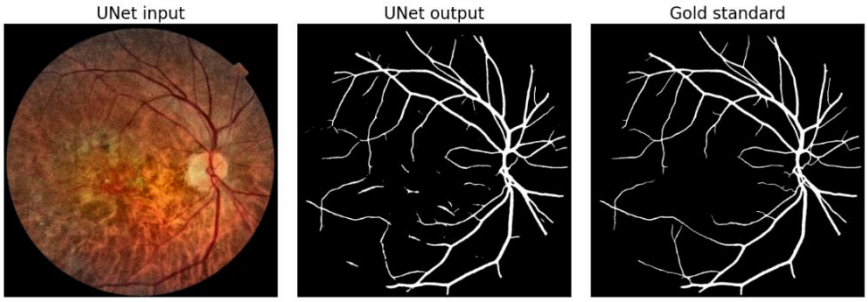



*Figure 7: Pre-processed image and its relative mask compared to the manual mask (Gold standard)*

## 2.6 Post-processing

Visual inspection of the results revealed two main critical issues: vascular discontinuity, characterized by interruptions in segments that should be continuous, and the presence of background noise, consisting of isolated pixels incorrectly classified as vessels. To resolve these issues the techniques listed in Table 6 have been tested. The most effective one turned out to be remove small objects (Figure 8) with a threshold of 60. However, this technique only acts by eliminating some of the FP and does not affect the reconstruction of the vessel. In addition, the chosen threshold doesn't remove large pieces of the vessel detected by the network that are actually FP, but increasing it would eliminate TP.

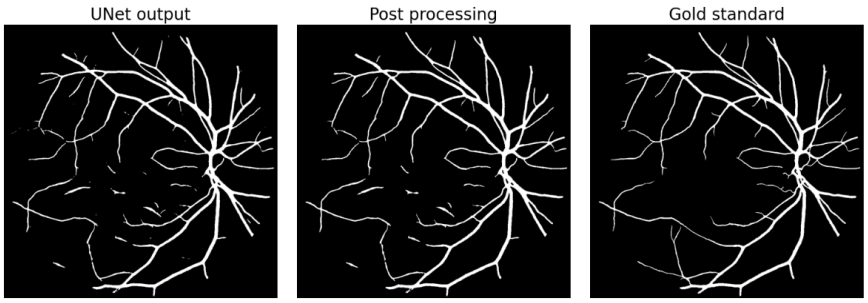



*Figure 8: Effect of post-processing on the image compared to the manual segmentation (Gold standard)*

## 2.7 Skeleton

A skeleton is a lower dimensional object that essentially represents the shape of its target object.
The process of generating such a skeleton is called skeletonization and the Python skeleton function is used to implement it.
Skeletonization reduces binary objects to 1-pixel wide representations by making successive passes of the image (Figure 9). On each pass, border pixels are identified and removed on the condition that they do not break the connectivity of the corresponding object [7].

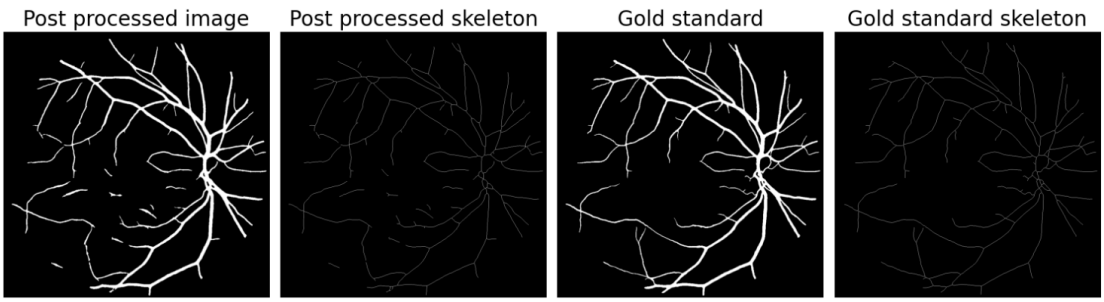



*Figure 9: Comparison of post-processed segmentation and vessel skeletonization*

# 3. RESULTS OBTAINED

The results that led to the final pipeline described in section 2 are presented in the below ablation study. To establish a performance benchmark, the baseline model was trained using raw RGB images without any preprocessing steps. Furthermore, the performance metrics reported in the subsequent tables for the pipeline optimization (including preprocessing, postprocessing and network parameters) refer to the validation set.

In the domain of retinal vessel segmentation, the extraction of the green channel is widely regarded as a standard preprocessing step. Extensive literature suggests that this channel typically provides the highest contrast between blood vessels and the background [8] [9].

Consequently, a specific experimental phase was conducted to evaluate the efficacy of the green channel within the proposed pipeline. Multiple configurations were tested (Table 1), combining Green Layer extraction with Color Normalization and various enhancement techniques, including CLAHE with different clip limits and tile sizes, as well as Gaussian, Average, and Median smoothing filters, to optimize performance.

Despite extensive parameter tuning, the single-channel approach consistently failed to match the results obtained using the full RGB information.

| IoU | Dice | Precision | Recall | Accuracy | MCC |
|---|---|---|---|---|---|
| **BASELINE** | | | | | |
| 0,692 ± 0,003 | 0,807 ± 0,129 | 0,903 ± 0,073 | 0,744 ± 0,154 | 0,987 ± 0,005 | 0,804 ± 0,117 |
| **COLOR NORMALIZATION (CN)+ GREEN-LAYER** | | | | | |
| 0,507 ± 0,144 | 0,651 ± 0,147 | 0,895 ± 0,121 | 0,539 ± 0,160 | 0,967 ± 0,008 | 0,668 ± 0,127 |
| **CN+ GREEN-LAYER+ CLAHE (clip limit= 2; 8x8)** | | | | | |
| 0,531 ± 0,144 | 0,680 ± 0,148 | 0,912 ± 0,098 | 0,557 ± 0,155 | 0,967 ± 0,007 | 0,691 ± 0,130 |
| **CN + GREEN-LAYER+ CLAHE (clip limit= 2.5; 16x16)** | | | | | |
| 0,551 ± 0,137 | 0,698 ± 0,140 | 0,831 ± 0,102 | 0,619 ± 0,163 | 0,966 ± 0,006 | 0,694 ± 0,126 |
| **CN+ GREEN-LAYER+ CLAHE (clip limit= 2.5; 16x16)+ GAUSSIAN (3x3 sigma=0.5)** | | | | | |
| 0,545 ± 0,147 | 0,691 ± 0,151 | 0,863 ± 0,104 | 0,596 ± 0,169 | 0,966 ± 0,007 | 0,693 ± 0,133 |
| **CN+ GREEN-LAYER+ CLAHE (clip limit= 2.5; 16x16)+ AVERAGE (size 3)** | | | | | |
| 0,547 ± 0,144 | 0,694 ± 0,149 | 0,887 ± 0,110 | 0,587 ± 0,162 | 0,967 ± 0,007 | 0,699 ± 0,132 |
| **CN+ GREEN-LAYER+ CLAHE (clip limit= 2.5; 16x16)+ MEDIAN (size 3)** | | | | | |
| 0,490 ± 0,139 | 0,645 ± 0,147 | 0,871 ± 0,122 | 0,527 ± 0,151 | 0,963 ± 0,008 | 0,654 ± 0,131 |

*Table 1: comparison between different strategies obtained with 10 epochs focused on the use of green layer*

Starting from the RGB baseline, the initial step of Color Normalization yielded a slight performance increase (Table 2), confirming its utility in standardizing the dataset. Subsequently, CLAHE was applied to enhance local contrast; extensive testing identified a Clip Limit of 2.5 and a Tile Grid Size of 16×16 as the optimal configuration. This setup achieved the highest metrics (e.g. Dice Score 0.813, MCC 0.807), providing the best balance between vessel–background contrast and image stability compared to other parameter combinations. As expected, this enhancement introduces a trade-off: while the improved Recall demonstrates the network's superior ability to detect thin vessels (reducing FN), the corresponding drop in Precision reflects an increase in FP, driven by the amplification of background noise that the network misinterprets as vascular structures.

| | IoU | Dice | Precision | Recall | Accuracy | MCC |
|---|---|---|---|---|---|---|
| **Baseline** | 0,692 ± 0,003 | 0,807 ± 0,129 | 0,903 ± 0,073 | 0,744 ± 0,154 | 0,987 ± 0,005 | 0,804 ± 0,117 |
| **Normalization** | 0,694 ± 0,152 | 0,808 ± 0,135 | 0,908 ± 0,067 | 0,746 ± 0,168 | 0,979 ± 0,006 | 0,806 ± 0,121 |
| **CLAHE (cl=3; 8x8)** | 0,652 ± 0,162 | 0,774 ± 0,155 | 0,913 ± 0,074 | 0,693 ± 0,176 | 0,976 ± 0,006 | 0,775 ± 0,139 |
| **CLAHE (cl=2,5; 12x12)** | 0,692 ± 0,145 | 0,807 ± 0,136 | 0,892 ± 0,107 | 0,748 ± 0,155 | 0,978 ± 0,005 | 0,802 ± 0,127 |
| **CLAHE (cl=2,5; 8x8)** | 0,691 ± 0,154 | 0,805 ± 0,142 | 0,897 ± 0,104 | 0,744 ± 0,167 | 0,979 ± 0,005 | 0,801 ± 0,131 |
| **CLAHE (cl=2,5; 16x16)** | 0,700 ± 0,142 | 0,813 ± 0,131 | 0,867 ± 0,108 | 0,778 ± 0,157 | 0,978 ± 0,005 | 0,807 ± 0,123 |
| **CLAHE (cl=2; 8x8)** | 0,686 ± 0,152 | 0,801 ± 0,141 | 0,881 ± 0,103 | 0,751 ± 0,169 | 0,978 ± 0,005 | 0,797 ± 0,131 |

*Table 2: comparison between different clip limits and tiles of CLAHE after normalization considering 10 epochs*

Although the optimized CLAHE configuration (Clip Limit 2.5, Grid Size 16x16) improved vessel contrast, it inevitably amplified high-frequency noise, manifesting as a granular texture in the background. To address this issue without compromising the enhanced vascular details, the final step of the ablation study focused on integrating a low-pass filtering stage.

The impact of three different smoothing kernels was evaluated after the CLAHE: Gaussian filter, Average filter, and Median filter. As presented in Table 3, the Gaussian filter with a 3x3 kernel and sigma=0.5 proved to be the most effective solution.

This specific configuration achieved a Dice Score of 0.817 and MCC 0.810; these results indicate that a mild Gaussian smoothing strikes an effective balance between noise reduction and preservation of thin vessel structures, leading to improved segmentation metrics compared to the unfiltered CLAHE baseline. This can be also seen in the increase of Precision, which reflects the suppression of noise-induced FP. The Averaging filter provided similar but slightly lower performance, while the Median filter resulted in a more noticeable drop, likely due to excessive smoothing of fine vascular details.

| | IoU | Dice | Precision | Recall | Accuracy | MCC |
|---|---|---|---|---|---|---|
| **Gaussian (sigma= 0,5; 3x3)** | 0,705 ± 0,140 | 0,817 ± 0,127 | 0,883 ± 0,085 | 0,771 ± 0,145 | 0,979 ± 0,005 | 0,810 ± 0,119 |
| **Gaussian (sigma= 1,0; 5x5)** | 0,683 ± 0,144 | 0,800 ± 0,136 | 0,844 ± 0,111 | 0,771 ± 0,162 | 0,977 ± 0,005 | 0,792 ± 0,130 |
| **Averaging (3x3)** | 0,702 ± 0,140 | 0,814 ± 0,126 | 0,867 ± 0,087 | 0,781 ± 0,152 | 0,978 ± 0,005 | 0,808 ± 0,118 |
| **Median (3x3)** | 0,675 ± 0,142 | 0,795 ± 0,130 | 0,849 ± 0,095 | 0,760 ± 0,160 | 0,976 ± 0,005 | 0,787 ± 0,122 |

*Table 3: comparison between different low pass filters after normalization and CLAHE (cl=2,5; 16x16) with 10 epochs*

Following the optimization of the preprocessing pipeline, the experimental focus shifted to the network training hyperparameters, specifically the number of the epochs was extended from 10 to 20.

Within this extended training regime, the preprocessing components that had proven most effective in previous steps were maintained: Color Normalization, CLAHE and Gaussian Smoothing. With this fixed pipeline structure, a re-optimization of the CLAHE parameters was conducted to verify if the optimal settings identified at 10 epochs remained valid under the new training conditions.

As presented in Table 4, the configuration with a Clip Limit of 2.5 and a Tile Grid Size of 16x16 once again emerged as the optimal choice demonstrating that the chosen preprocessing parameters are robust and independent from the number of epochs. This setup, combined with the extended training, achieved a superior Dice Score of 0.832 and MCC 0.826; this increase demonstrates that the network benefited significantly from the additional epochs to refine its convergence on complex vascular structures.

| | IoU | Dice | Precision | Recall | Accuracy | MCC |
|---|---|---|---|---|---|---|
| **Baseline** | 0,694 ± 0,152 | 0,808 ± 0,135 | 0,908 ± 0,067 | 0,746 ± 0,168 | 0,979 ± 0,006 | 0,806 ± 0,121 |
| **CLAHE (cl=3; 8x8)** | 0,722 ± 0,136 | 0,829 ± 0,123 | 0,886 ± 0,091 | 0,789 ± 0,145 | 0,980 ± 0,005 | 0,823 ± 0,116 |
| **CLAHE (cl=2,5; 12x12)** | 0,725 ± 0,138 | 0,831 ± 0,127 | 0,869 ± 0,094 | 0,806 ± 0,152 | 0,980 ± 0,004 | 0,824 ± 0,119 |
| **CLAHE (cl=2,5; 8x8)** | 0,715 ± 0,141 | 0,824 ± 0,129 | 0,905 ± 0,099 | 0,765 ± 0,148 | 0,980 ± 0,005 | 0,819 ± 0,122 |
| **CLAHE (cl=2,5; 16x16)** | 0,726 ± 0,135 | 0,832 ± 0,121 | 0,870 ± 0,070 | 0,810 ± 0,148 | 0,980 ± 0,004 | 0,826 ± 0,111 |
| **CLAHE (cl=2; 8x8)** | 0,711 ± 0,142 | 0,820 ± 0,130 | 0,887 ± 0,082 | 0,778 ± 0,155 | 0,980 ± 0,005 | 0,815 ± 0,120 |

*Table 4: Comparison between different clip limits and tiles of CLAHE with normalization and Gaussian (sigma=0,5;3x3) considering 20 epochs*

In addition to tuning the training epochs, the optimization of the batch size was explored to further improve model performance.

However, attempts to increase the batch size beyond 4 keeping the native resolution (1024x1024) triggered CUDA OutOfMemory errors, exceeding the available GPU capacity. To overcome this and enable training with larger batches, we evaluated a resizing strategy, resizing the input images to 512x512.

As presented in Table 5, this trade-off revealed a decrease of the segmentation performance. The model trained on resized images recorded a significant drop in evaluation metrics, such as Dice of 0.785 and MCC of 0.783. This degradation indicates that resizing causes a loss of fine-grained spatial details required to detect retinal vessels. Consequently, the resizing strategy was discarded, prioritizing the preservation of original image details over a larger batch size.

| | IoU | Dice | Precision | Recall | Accuracy | MCC |
|---|---|---|---|---|---|---|
| **Baseline** | 0,694 ± 0,152 | 0,808 ± 0,135 | 0,908 ± 0,067 | 0,746 ± 0,168 | 0,979 ± 0,006 | 0,806 ± 0,121 |
| **Resize (512x512)** | 0,658 ± 0,124 | 0,785 ± 0,119 | 0,918 ± 0,061 | 0,697 ± 0,132 | 0,976 ± 0,004 | 0,783 ± 0,109 |

*Table 5: Comparison between the original images changing the batch size using 20 epochs*

For the final post-processing strategy, the Remove Small Objects filter was applied (with a threshold of 60 pixels). This method proved to be the only step capable of enhancing performance metrics achieving a Dice score of 0.833 and an MCC of 0.827 by eliminating background FP without introducing structural artifacts. This decision followed the ineffectiveness of the techniques explored for reconnecting fragmented vessels. Initially, morphological operators were evaluated, prioritizing Closing over Opening to avoid the risk of eroding thin terminal vessels associated with the latter. However, the Closing operation erroneously merged close vessels. Subsequently, the Local-Sensitive Connectivity Filter (LS-CF) [10] was tested as a more selective approach, searching for connections only near vessel endpoints (within a 10-pixel range). Since even varying the search distance failed to yield an effective trade-off, these reconnection strategies were ultimately discarded. In Table 6 are shown the results obtained after optimizing the parameters for each technique.

| | IoU | Dice | Precision | Recall | Accuracy | MCC |
|---|---|---|---|---|---|---|
| **Closing+ Remove Small Objects** | 0,726 ± 0,134 | 0,832 ± 0,120 | 0,870 ± 0,069 | 0,811 ± 0,148 | 0,980 ± 0,004 | 0,826 ± 0,110 |
| **Remove Small Objects** | 0,728 ± 0,136 | 0,833 ± 0,122 | 0,876 ± 0,068 | 0,807 ± 0,148 | 0,980 ± 0,004 | 0,827 ± 0,111 |
| **LSCF** | 0,726 ± 0,134 | 0,832 ± 0,120 | 0,868 ± 0,070 | 0,811 ± 0,147 | 0,980 ± 0,004 | 0,825 ± 0,111 |

*Table 6: Comparison between different strategies of post processing*

# 4. DISCUSSION AND CONCLUSION

A critical insight gained during the ablation study was the superiority of the full RGB information over the traditional green-channel extraction. While literature often favors the green channel for its contrast, our findings suggest that the integration of color normalization and CLAHE on the original RGB image preserves essential vascular details that are lost in single-channel approaches. However, a persistent challenge remains the structural fragmentation of thinner vessels, which the model occasionally fails to represent as continuous segments. Efforts to resolve this via morphological closing or local connectivity filters were ultimately unsuccessful, as they tended to erroneously merge close vessels rather than accurately reconnecting broken ones.

A significant portion of the development process was defined by the inherent trade-off between image resolution and hardware capacity. To preserve the fine-grained spatial details necessary for detecting thin vessels, the native 1024x1024 resolution was prioritized. This choice, however, reached the physical limits of the available hardware, as attempts to increase the batch size or deepen the network architecture consistently triggered CUDA OutOfMemory errors. While reducing the resolution to 512x512 allowed for larger batch sizes, it led to a significant degradation in segmentation performance, confirming that spatial detail is more critical than batch optimization for this task. It is highly probable that with superior GPU resources, that could allow for the simultaneous use of high resolution and larger batch sizes or more complex architecture, the model's convergence and accuracy could be further improved.

Considering the experimental evidence gathered from the ablation study on the validation set, the final preprocessing pipeline was ROI extraction, Color Normalization, CLAHE (Clip Limit 2.5, Grid Size 16x16), and Gaussian Filtering (sigma=0.5). To further refine the segmentation output, a post-processing step based on the Remove Small Objects technique was introduced. With this optimized configuration fixed, the network was ultimately subjected to evaluation on the Test Set to assess its generalization capabilities on unseen data (Table 7). The consistency between validation and test metrics suggests that the pipeline is robust; however, future work could explore sophisticated post-processing techniques aimed at improving vessel connectivity and mitigating over-segmentation.

| | IoU | Dice | Precision | Recall | Accuracy | MCC | clDice |
|---|---|---|---|---|---|---|---|
| TEST SET | 0,715 ± 0,142 | 0,823 ± 0,131 | 0,857 ± 0,127 | 0,799 ± 0,151 | 0,980 ± 0,006 | 0,815 ± 0,130 | 0,824 ± 0,140 |
| VALIDATION SET | 0,728 ± 0,136 | 0,833 ± 0,122 | 0,876 ± 0,068 | 0,807 ± 0,148 | 0,980 ± 0,004 | 0,827 ± 0,111 | 0,834 ± 0,131 |

*Table 7: performance of the final pipeline on validation and test set*

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